

MIRCO MUTTI

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RESEARCH INTERESTS

Reinforcement learning, theoretical and methodological foundations for real-world deployment including: Exploration; Generalization; Partial observability; General utilities; Interpretability

EMPLOYMENT

Postdoctoral Researcher Technion , Haifa, Israel Advisor: Aviv Tamar Research: Foundations of practical reinforcement learning	2023–
Research Fellow Politecnico di Milano , Milan, Italy Advisor: Marcello Restelli	2018

EDUCATION

PhD , Data Science and Computation Politecnico di Milano & Università di Bologna , Milan, Italy Supervisor: Marcello Restelli Thesis: Unsupervised reinforcement learning via state entropy maximization	2018-2023
MSc , Computer Science and Engineering (110/110 cum laude) Politecnico di Milano , Milan, Italy Supervisor: Marcello Restelli Thesis: Configurable Markov decision processes	2015-2018
BSc , Engineering of Computing Systems (110/110 cum laude) Politecnico di Milano , Milan, Italy	2012-2015

HONORS & AWARDS

Oral presentation at NeurIPS 2025	2025
Best paper award at the Exploration in AI Today workshop at ICML 2025	2025
Honorable mention for the best PhD thesis on artificial intelligence by AIXIA	2023
Notable paper at AISTATS 2023	2023
Outstanding paper award at ICML 2022	2022
Best thesis award for Master's thesis on artificial intelligence by AIXIA	2019
Ph.D. scholarship for merits by the Italian ministry of university and research	2018
Expert reviewer for the TMLR journal '22, '23, '24	
Outstanding reviewer at NeurIPS '21, '22, '23, '24, '25, ICLR '22, '24, ICML '22, '24	

SERVICE & PEER REVIEW

Program Co-Chair of the European Workshop on Reinforcement Learning (EWRL 2022)

Action Editor of the TMLR journal 2023, 2024, 2025

Reviewer of

NeurIPS **2021**^{*}, **2022**^{*}, **2023**^{*}, **2024**^{*}, **2025**^{*}

^{*}Outstanding reviewer

ICML 2021, **2022**^{*}, 2023, **2024**^{*}, 2025

ICLR 2021, **2022**^{*}, 2023, **2024**^{*}, 2025, 2026

RLC 2024, 2025

AAAI 2021, 2022, 2023

AISTATS 2021, 2022

IJCAI 2020, 2021

TMLR **2022**[†], **2023**[†], **2024**[†]

[†]Expert reviewer

Machine Learning, JAIR, Neural Networks, IEEE RA-L

TEACHING

Machine learning, Politecnico di Milano, TA (10h of exercise sessions $\approx$ 400 students) Spring 2022

Informatics B, Politecnico di Milano, TA (26h of exercise sessions $\approx$ 200 students) Fall 2021

Machine learning, Politecnico di Milano, TA (18h of exercise sessions $\approx$ 400 students) Spring 2021

Informatics B, Politecnico di Milano, TA (36h of exercise sessions $\approx$ 200 students) Fall 2020

Machine learning, Politecnico di Milano, TA (10h of exercise sessions $\approx$ 400 students) Spring 2020

Informatics B, Politecnico di Milano, TA (26h of exercise sessions $\approx$ 200 students) Fall 2019

Informatics B, Politecnico di Milano, TA (26h of exercise sessions $\approx$ 200 students) Fall 2018

INVITED TALKS

Fundamentals of reinforcement learning generalization Nov 2025

Princeton RL lab (host: Benjamin Eysenbach)

Theoretical considerations for practical RL May 2025

University of Verona (host: Alessandro Farinelli)

Bocconi University - Milan (host: Andrea Celli)

CSML lab at IIT - Genova (host: Massimiliano Pontil)

MaLGa center and University of Genoa (host: Lorenzo Rosasco)

A classification view on meta learning bandits Apr 2025

Theory of Interpretable AI Seminar

(Non)Convex reinforcement learning

LAS group & AI Center at ETH Zurich (host: Andreas Krause) Jun 2023

Rising Stars in AI Symposium 2023 at KAUST Feb 2023

Unsupervised reinforcement learning

VANDAL lab at Politecnico di Torino (host: Raffaello Camoriano) Apr 2024

Robot Learning Lab at the Technion (host: Aviv Tamar) Sep 2022

Facebook AI Research - External Seminars May 2022

Configurable Markov decision processes Nov 2019

AixIA Conference - Cosenza

MENTORSHIP (CO-SUPERVISION)

Omer Hemo , MSc ECE Technion	2025-
Yonatan Ashlag , MSc ECE Technion (publication at NeurIPS)	2024-2025
Duilio Cirino , MSc CE Politecnico di Milano (publications at ICML, RLC)	2023-2024
Stefano Civelli , MSc CE Politecnico di Milano	2023
Vlad Marian Cimpeanu , MSc CE Politecnico di Milano	2022-2023
Majid Molaei , MSc TE Politecnico di Milano	2021-2023
Juan Calderon , MSc CE Politecnico di Milano (publication at AAAI)	2021-2022
Pietro Maldini , MSc CE Politecnico di Milano	2021-2022
Mattia Mancassola , MSc CE Politecnico di Milano (publication at AAAI)	2020-2021
Stefano Del Col , MSc CE Politecnico di Milano (publication at AISTATS)	2020-2021
Riccardo De Santi , MSc CS ETH Zurich (publications at ICML, NeurIPS, ICLR, AAAI)	2019-2023
Lorenzo Pratissoli , MSc CE Politecnico di Milano (publication at AAAI)	2019-2020
Giovanni Pelosi , MSc CE Politecnico di Milano	2018-2019

PUBLICATIONS

Publications: 24; citations: 480; h-index: 10

—Journal—

- **Convex reinforcement learning in finite trials**
Mirco Mutti, Riccardo De Santi, Piersilvio De Bartolomeis, Marcello Restelli
(JMLR'23) *Journal of Machine Learning Research*

—Conference proceedings—

23. **State entropy regularization for robust reinforcement learning**
Yonatan Ashlag, Uri Koren, Mirco Mutti, Esther Derman, Pierre-Luc Bacon, Shie Mannor
(NeurIPS'25) *Neural Information Processing Systems*, with *oral presentation* (< 1%)
22. **Blindfolded experts generalize better: Insights from robotic manipulation and videogames**
Ev Zisselman, Mirco Mutti, Shelly Francis-Meretzki, Elisei Shefer, Aviv Tamar
(NeurIPS'25) *Neural Information Processing Systems*
21. **Towards principled unsupervised multi-agent reinforcement learning**
Riccardo Zamboni, Mirco Mutti, Marcello Restelli
(NeurIPS'25) *Neural Information Processing Systems*
20. **Enhancing diversity in parallel agents: A maximum state entropy exploration story**
Vincenzo De Paola, Riccardo Zamboni, Mirco Mutti, Marcello Restelli
(ICML'25) *International Conference on Machine Learning*
19. **A classification view on meta learning bandits**
Mirco Mutti, Jeongyeol Kwon, Shie Mannor, Aviv Tamar
(ICML'25) *International Conference on Machine Learning*
18. **A theoretical framework for partially-observed reward-states in RLHF**
Chinmaya Kausik, Mirco Mutti, Aldo Pacchiano, Ambuj Tewari
(ICLR'25) *International Conference on Learning Representations*

17. **How does inverse RL scale to large state spaces? A provably efficient approach**
Filippo Lazzati, [Mirco Mutti](#), Alberto Maria Metelli
(NeurIPS'24) *Neural Information Processing Systems*
16. **The limits of pure exploration in POMDPs: When the observation entropy is enough**
Riccardo Zamboni, Duilio Cirino, Marcello Restelli, [Mirco Mutti](#)
(RLC'24) *Reinforcement Learning Conference*
15. **How to explore with belief: State entropy maximization in POMDPs**
Riccardo Zamboni, Duilio Cirino, Marcello Restelli, [Mirco Mutti](#)
(ICML'24) *International Conference on Machine Learning*
14. **Test-time regret minimization in meta reinforcement learning**
[Mirco Mutti](#), Aviv Tamar
(ICML'24) *International Conference on Machine Learning*
13. **Geometric active exploration in Markov decision processes: The benefit of abstraction**
Riccardo De Santi, Federico Arangath Joseph, Noah Liniger, [Mirco Mutti](#), Andreas Krause
(ICML'24) *International Conference on Machine Learning*
12. **Offline inverse RL: New solution concepts and provably efficient algorithms**
Filippo Lazzati, [Mirco Mutti](#), Alberto Maria Metelli
(ICML'24) *International Conference on Machine Learning*
11. **Exploiting causal graph priors with posterior sampling for reinforcement learning**
[Mirco Mutti](#), Riccardo De Santi, Marcello Restelli, Alexander Marx, Giorgia Ramponi
(ICLR'24) *International Conference on Learning Representations*
10. **Persuading farsighted receivers in MDPs: The power of honesty**
Martino Bernasconi, Matteo Castiglioni, Alberto Marchesi, [Mirco Mutti](#)
(NeurIPS'23) *Neural Information Processing Systems*
9. **A tale of sampling and estimation in discounted reinforcement learning**
Alberto Maria Metelli, [Mirco Mutti](#), Marcello Restelli
(AISTATS'23) *Conference on Artificial Intelligence and Statistics*, *notable paper* (top 2%)
8. **Provably efficient causal model-based reinforcement learning for systematic generalization** [Mirco Mutti](#), Riccardo De Santi, Emanuele Rossi, Juan Felipe Calderon, Michael Bronstein, Marcello Restelli
(AAAI'23) *Association for the Advancements of Artificial Intelligence*
7. **Challenging common assumptions in convex reinforcement learning**
[Mirco Mutti](#), Riccardo De Santi, Piersilvio De Bartolomeis, Marcello Restelli
(NeurIPS'22) *Neural Information Processing Systems*
6. **The importance of non-Markovianity in maximum state entropy exploration**
[Mirco Mutti](#), Riccardo De Santi, Marcello Restelli
(ICML'22) *International Conference on Machine Learning*, *outstanding paper award*
5. **Reward-free policy space compression for reinforcement learning**
[Mirco Mutti](#), Stefano Del Col, Marcello Restelli
(AISTATS'22) *Conference on Artificial Intelligence and Statistics*

4. **Unsupervised reinforcement learning in multiple environments**
Mirco Mutti, Mattia Mancassola, Marcello Restelli
(AAAI'22) *Association for the Advancements of Artificial Intelligence*
3. **Task-agnostic exploration via policy gradient of a non-parametric state entropy estimate**
Mirco Mutti, Lorenzo Pratissoli, Marcello Restelli
(AAAI'21) *Association for the Advancements of Artificial Intelligence*
2. **An intrinsically-motivated approach for learning highly exploring and fast mixing policies**
Mirco Mutti, Marcello Restelli
(AAAI'20) *Association for the Advancements of Artificial Intelligence*
1. **Configurable Markov decision processes**
Alberto Maria Metelli, Mirco Mutti, Marcello Restelli
(ICML'18) *International Conference on Machine Learning, with **long talk*** (top 8.5%)

—Preprint—

- **Reward compatibility: A framework for inverse RL**
Filippo Lazzati, Mirco Mutti, Alberto Maria Metelli
(ArXiv 2025)

—Selected Workshop—

- **Blindfolded experts generalize better: Insights from robotic manipulation and videogames**
Ev Zisselman, Mirco Mutti, Shelly Francis-Meretzki, Elisei Shefer, Aviv Tamar
(EXAIT @ ICML'25) *Workshop on Exploration in AI Today, **best paper award***
- **Exploiting causal graph priors with posterior sampling for reinforcement learning**
Mirco Mutti, Riccardo De Santi, Marcello Restelli, Alexander Marx, Giorgia Ramponi
(EWRL'23) *European Workshop on Reinforcement Learning*
- **Provably efficient causal model-based reinforcement learning for systematic generalization** Mirco Mutti, Riccardo De Santi, Emanuele Rossi, Juan Felipe Calderon, Michael Bronstein, Marcello Restelli
(RLDM'22) *Workshop on Reinforcement Learning and Decision Making*
- **An approach for learning highly exploring and fast mixing policies via intrinsic motivation**
Mirco Mutti, Marcello Restelli
(IMOL'19) *Workshop on Intrinsically Motivated Open-ended Learning*
- **Configurable Markov decision processes**
Alberto Maria Metelli, Mirco Mutti, Marcello Restelli
(EWRL'18) *European Workshop on Reinforcement Learning*